20XW97-COMPUTER VISION ABSTRACT

Detection and Segmentation of Cement Concrete Pavement Pothole Based on Image Processing Technology

Pothole detection and segmentation are important for automated pavement condition assessment and road safety. Existing image-processing-based approaches have demonstrated the effectiveness of texture and grayscale features for identifying potholes in cement concrete pavements. However, potholes with weak texture, low contrast, sandy or irregular surface materials, and unclear boundaries can be difficult to segment accurately using texture information alone.

This study proposes an enhanced pothole detection and segmentation approach by incorporating edge information with the original texture-based representation. Edge features are extracted from the input pavement image and integrated with the texture information to strengthen the structural and boundary characteristics of potholes. The combined texture-edge representation is subsequently processed using grayscale conversion, thresholding, and morphological operations to obtain a refined pothole region.

The proposed approach aims to preserve the texture-based ability to distinguish potholes from the surrounding pavement while improving the localization of weak or poorly defined pothole boundaries. The performance of the enhanced method will be evaluated against the original texture-based approach using segmentation and detection metrics such as precision, recall, F1-score, and overlap/IoU. The proposed enhancement is expected to provide more reliable pothole segmentation, particularly in challenging pavement conditions where texture information alone is insufficient.

APPROACH :

Texture + Edge information → Enhanced representation → Grayscale → Binary → Morphology → Pothole

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